

ELITE MATHEMATICS

Pearson Edexcel International A Level

Mechanics 1

Adaptive Classified

Chapter 02 · Working with Vectors

Questions

Student book · questions with writing space

- 94 pages
- Past papers up to June 2026

MECHANICS 1 | TOPIC 02

Working with Vectors

Your classified route

30

QUESTIONS

323

MARKS

4

FAMILIES

11

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Read the diagram, choose the model, then use the working space to complete the mechanics.

TOPIC 02 | FAMILY 01

Resolve vectors and form resultants

WHAT THIS FAMILY ASKS

Convert magnitudes/directions into i-j components, add vectors and recover magnitude or direction.

VARIANT ROUTE

V01 Bearing or angle to components

V02 Decompose into prescribed parallel directions

V03 Resultant carried into dynamics

V04 Unknown components from a required resultant direction

V05 Non-component force triangle

10 classified questions

85 total marks

7. Two forces, $\mathbf{F}$ and $\mathbf{G}$, act on a particle. The force $\mathbf{F}$ has magnitude 4 N and acts in a direction with a bearing of 120° and the force $\mathbf{G}$ has magnitude 6 N and acts due north.

Given that $\mathbf{P} = 2\mathbf{F} + \mathbf{G}$, find

- (i) the magnitude of $\mathbf{P}$
- (ii) the direction of $\mathbf{P}$, giving your answer as a bearing to the nearest degree.

(7)

DR ESLAM AHMED

5. A particle is acted upon by two forces $\mathbf{F}$ and $\mathbf{G}$. The force $\mathbf{F}$ has magnitude 8 N and acts in a direction with a bearing of 240° . The force $\mathbf{G}$ has magnitude 10 N and acts due South.

Given that $\mathbf{R} = \mathbf{F} + \mathbf{G}$, find

- (i) the magnitude of $\mathbf{R}$,
- (ii) the direction of $\mathbf{R}$, giving your answer as a bearing to the nearest degree.

(7)

DR ESLAM AHMED

2. Two forces, $\mathbf{P}$ and $\mathbf{Q}$, act on a particle.

- $\mathbf{P}$ has magnitude 10 N and acts due west
- $\mathbf{Q}$ has magnitude 8 N and acts on a bearing of 330°

Given that $\mathbf{F} = \mathbf{P} + \mathbf{Q}$, find the magnitude of $\mathbf{F}$.

(4)

DR ESAM AHMED

6. The resultant $\mathbf{R}$ of two forces $\mathbf{F}_1 \text{ N}$ and $\mathbf{F}_2 \text{ N}$ is $(-5\mathbf{i} + 6\mathbf{j}) \text{ N}$.

(a) Find the magnitude of $\mathbf{R}$.

(2)

DR ESAM AHMED

6. The resultant $\mathbf{R}$ of two forces $\mathbf{F}_1 \text{ N}$ and $\mathbf{F}_2 \text{ N}$ is $(-5\mathbf{i} + 6\mathbf{j}) \text{ N}$.

- (b) Find, in degrees, the angle between the direction of $\mathbf{R}$ and the direction of the vector $(2\mathbf{i} + \mathbf{j})$. Show your method clearly.

(3)

DR ESAM AHMED

6. The resultant $\mathbf{R}$ of two forces $\mathbf{F}_1 \text{ N}$ and $\mathbf{F}_2 \text{ N}$ is $(-5\mathbf{i} + 6\mathbf{j}) \text{ N}$.

Given that $\mathbf{F}_1$ is parallel to the vector $(-\mathbf{i} + 2\mathbf{j})$ and that $\mathbf{F}_2$ is parallel to the vector $(2\mathbf{i} - 3\mathbf{j})$,

- (c) find $\mathbf{F}_1$ and $\mathbf{F}_2$

(4)

DR ESLAM AHMED

3. [In this question $\mathbf{i}$ and $\mathbf{j}$ are perpendicular horizontal unit vectors.]

Three forces, $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, are given by

$$\mathbf{F}_1 = (5\mathbf{i} + 2\mathbf{j})\text{N} \quad \mathbf{F}_2 = (-3\mathbf{i} + \mathbf{j})\text{N} \quad \mathbf{F}_3 = (a\mathbf{i} + b\mathbf{j})\text{N}$$

where a and b are constants.

The forces $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ act on a particle P of mass 4 kg.

Given that P rests in equilibrium on a smooth horizontal surface under the action of these three forces,

- (a) find the size of the angle between the direction of $\mathbf{F}_3$ and the direction of $-\mathbf{j}$.

(4)

DR ESAM AHMED

3. [In this question $\mathbf{i}$ and $\mathbf{j}$ are perpendicular horizontal unit vectors.]

Three forces, $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, are given by

$$\mathbf{F}_1 = (5\mathbf{i} + 2\mathbf{j})\text{N} \quad \mathbf{F}_2 = (-3\mathbf{i} + \mathbf{j})\text{N} \quad \mathbf{F}_3 = (a\mathbf{i} + b\mathbf{j})\text{N}$$

where a and b are constants.

The forces $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ act on a particle P of mass 4 kg.

The force $\mathbf{F}_3$ is now removed and replaced by the force $\mathbf{F}_4$ given by $\mathbf{F}_4 = \lambda(\mathbf{i} + 3\mathbf{j})\text{N}$, where λ is a positive constant.

When the three forces $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_4$ act on P , the acceleration of P has magnitude 3.25 ms^{-2}

(b) Find the value of λ .

(5)

DR ESAM AHMED

[In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.]

2. A particle P rests in equilibrium on a smooth horizontal plane.

A system of **three** forces, $\mathbf{F}_1$ N, $\mathbf{F}_2$ N and $\mathbf{F}_3$ N where

$$\mathbf{F}_1 = (3c\mathbf{i} + 4c\mathbf{j})$$

$$\mathbf{F}_2 = (-14\mathbf{i} + 7\mathbf{j})$$

is applied to P .

Given that P remains in equilibrium,

- (a) find $\mathbf{F}_3$ in terms of c , $\mathbf{i}$ and $\mathbf{j}$.

(2)

DR ESAM AHMED

[In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.]

2. A particle P rests in equilibrium on a smooth horizontal plane.

A system of **three** forces, $\mathbf{F}_1$ N, $\mathbf{F}_2$ N and $\mathbf{F}_3$ N where

$$\mathbf{F}_1 = (3c\mathbf{i} + 4c\mathbf{j})$$

$$\mathbf{F}_2 = (-14\mathbf{i} + 7\mathbf{j})$$

is applied to P .

The force $\mathbf{F}_3$ is **removed** from the system.

Given that $c = 2$

- (b) find the size of the angle between the direction of $\mathbf{i}$ and the direction of the resultant force acting on P .

(4)

DR ESAM AHMED

[In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.]

2. A particle P rests in equilibrium on a smooth horizontal plane.

A system of **three** forces, $\mathbf{F}_1$ N, $\mathbf{F}_2$ N and $\mathbf{F}_3$ N where

$$\mathbf{F}_1 = (3c\mathbf{i} + 4c\mathbf{j})$$

$$\mathbf{F}_2 = (-14\mathbf{i} + 7\mathbf{j})$$

is applied to P .

The force $\mathbf{F}_3$ is **removed** from the system.

Given that $c = 2$

The mass of P is m kg.

Given that the magnitude of the acceleration of P is 8.5 m s^{-2}

(c) find the value of m .

(4)

DR ESAM AHMED

3. [In this question **i** and **j** are horizontal perpendicular unit vectors.]

A particle *P* of mass 2 kg moves on a smooth horizontal surface under the action of two forces **F**₁ and **F**₂, where **F**₁ = (−2**i** + 3**j**)N and **F**₂ = (4**i** + 2**j**)N.

(a) Find the acceleration of *P*.

(3)

DR ESAM AHMED

3. [In this question **i** and **j** are horizontal perpendicular unit vectors.]

A particle *P* of mass 2 kg moves on a smooth horizontal surface under the action of two forces **F**₁ and **F**₂, where **F**₁ = (−2**i** + 3**j**)N and **F**₂ = (4**i** + 2**j**)N.

At time *t* = 0, the velocity of *P* is (3**i** − 4**j**) m s^{−1}

(b) Find the speed of *P* when *t* = 3 seconds.

(4)

DR ESAM AHMED

3. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.]

A particle P of mass 2 kg moves on a smooth horizontal surface under the action of two forces $\mathbf{F}_1$ and $\mathbf{F}_2$, where $\mathbf{F}_1 = (-2\mathbf{i} + 3\mathbf{j})\text{N}$ and $\mathbf{F}_2 = (4\mathbf{i} + 2\mathbf{j})\text{N}$.

An additional force, $\mathbf{F}_3 = (b\mathbf{i} + c\mathbf{j})\text{N}$, is applied to P .

The resultant of $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ is equal to $\lambda(\mathbf{i} + \mathbf{j})\text{N}$, where λ is a constant.

- (c) Show that $b - c = 3$

(3)

DR ESAM AHMED

3. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.]

A particle P of mass 2 kg moves on a smooth horizontal surface under the action of two forces $\mathbf{F}_1$ and $\mathbf{F}_2$, where $\mathbf{F}_1 = (-2\mathbf{i} + 3\mathbf{j})\text{N}$ and $\mathbf{F}_2 = (4\mathbf{i} + 2\mathbf{j})\text{N}$.

An additional force, $\mathbf{F}_3 = (b\mathbf{i} + c\mathbf{j})\text{N}$, is applied to P .

The resultant of $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ is equal to $\lambda(\mathbf{i} + \mathbf{j})\text{N}$, where λ is a constant.

The resultant of $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ has magnitude $10\sqrt{2}\text{N}$.

- (d) Find the two possible $\mathbf{F}_3$ forces.

(4)

DR ESAM AHMED

6. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle A of mass 0.5 kg is at rest on a smooth horizontal plane.

At time $t = 0$, two forces, $\mathbf{F}_1 = (-3\mathbf{i} + 2\mathbf{j})\text{ N}$ and $\mathbf{F}_2 = (p\mathbf{i} + q\mathbf{j})\text{ N}$, where p and q are constants, are applied to A .

Given that A moves in the direction of the vector $(\mathbf{i} - 2\mathbf{j})$,

(a) show that $2p + q - 4 = 0$

(4)

DR ESAM AHMED

6. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle A of mass 0.5 kg is at rest on a smooth horizontal plane.

At time $t = 0$, two forces, $\mathbf{F}_1 = (-3\mathbf{i} + 2\mathbf{j})\text{ N}$ and $\mathbf{F}_2 = (p\mathbf{i} + q\mathbf{j})\text{ N}$, where p and q are constants, are applied to A .

Given that $p = 5$

- (b) find the speed of A at time $t = 4$ seconds.

(5)

DR ESAM AHMED

1. A particle of mass 2.5 kg moves on a smooth horizontal plane under the action of three horizontal forces, $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, where

$$\mathbf{F}_1 = (6\mathbf{i} + 8\mathbf{j})\text{N}$$

$$\mathbf{F}_2 = (-16\mathbf{i} + 2\mathbf{j})\text{N}$$

$$\mathbf{F}_3 = (-2\mathbf{i} + 8\mathbf{j})\text{N}$$

- (a) Find the magnitude of the acceleration of the particle.

(4)

DR ESAM AHMED

The question currently reads:

- The resultant of the four forces acts in **the direction of** the vector $(7\mathbf{i} + 2\mathbf{j})$.

It should read:

- The resultant of the four forces acts in **a direction that is parallel to** the vector $(7\mathbf{i} + 2\mathbf{j})$.

1. A particle of mass 2.5 kg moves on a smooth horizontal plane under the action of three horizontal forces, $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, where

$$\mathbf{F}_1 = (6\mathbf{i} + 8\mathbf{j})\text{N}$$

$$\mathbf{F}_2 = (-16\mathbf{i} + 2\mathbf{j})\text{N}$$

$$\mathbf{F}_3 = (-2\mathbf{i} + 8\mathbf{j})\text{N}$$

A fourth force, $\mathbf{F}_4 = (p\mathbf{i} + p\mathbf{j})\text{N}$, where p is a constant, is added.

The resultant of the four forces acts in the direction of the vector $(7\mathbf{i} + 2\mathbf{j})$.

- (b) Find the value of p .

(4)

DR ESAM AHMED

4.

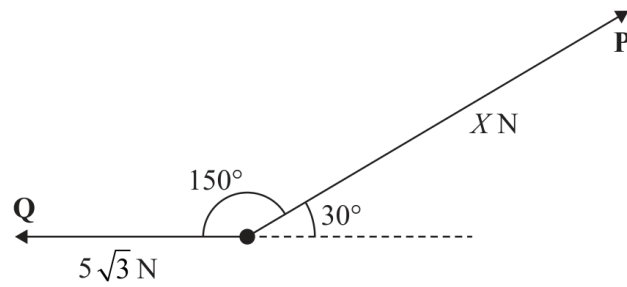
**Figure 3**

Figure 3 shows two horizontal forces **P** and **Q** acting on a particle.

The angle between the direction of **P** and the direction of **Q** is 150°

Force **P** has magnitude X newtons.

Force **Q** has magnitude $5\sqrt{3}$ N.

The resultant of **P** and **Q** has magnitude $\sqrt{129}$ N.

Find

- (i) the value of X .
- (ii) the angle between **Q** and the resultant, giving your answer to the nearest degree.

(8)

TOPIC 02 | FAMILY 02

Angle, bearing and direction constraints

WHAT THIS FAMILY ASKS

Determine or impose the direction of a vector using quadrants, bearings, angles, parallelism or a dot product.

VARIANT ROUTE

V01 Parallel, opposite or same-direction condition

4 classified questions

41 total marks

6. A force $\mathbf{F}$ is given by $\mathbf{F} = (10\mathbf{i} + \mathbf{j})\text{N}$.

(a) Find the exact value of the magnitude of $\mathbf{F}$.

(2)

DR ESLAM AHMED

6. A force $\mathbf{F}$ is given by $\mathbf{F} = (10\mathbf{i} + \mathbf{j})\text{N}$.
(b) Find, in degrees, the size of the angle between the direction of $\mathbf{F}$ and the direction of the vector $(\mathbf{i} + \mathbf{j})$.

(4)

DR ESAM AHMED

6. A force $\mathbf{F}$ is given by $\mathbf{F} = (10\mathbf{i} + \mathbf{j})\text{N}$.
The resultant of the force $\mathbf{F}$ and the force $(-15\mathbf{i} + a\mathbf{j})\text{N}$, where a is a constant, is parallel to, but in the opposite direction to, the vector $(2\mathbf{i} - 3\mathbf{j})$.

(c) Find the value of a .

(5)

DR ESAM AHMED

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle P of mass 2 kg moves under the action of two forces, $(p\mathbf{i} + q\mathbf{j})\text{ N}$ and $(2q\mathbf{i} + p\mathbf{j})\text{ N}$, where p and q are constants.

Given that the acceleration of P is $(\mathbf{i} - \mathbf{j})\text{ m s}^{-2}$

(a) find the value of p and the value of q .

(5)

DR ESAM AHMED

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

Given that the acceleration of P is $(\mathbf{i} - \mathbf{j}) \text{ m s}^{-2}$

(b) Find the size of the angle between the direction of the acceleration and the vector $\mathbf{j}$.
(2)

DR ESAM AHMED

Given that the acceleration of P is $(\mathbf{i} - \mathbf{j}) \text{ m s}^{-2}$

At time $t = 0$, the velocity of P is $(3\mathbf{i} - 4\mathbf{j}) \text{ m s}^{-1}$

At $t = T$ seconds, P is moving in the direction of the vector $(11\mathbf{i} - 13\mathbf{j})$.

(c) Find the value of T .

(5)

DR ESAM AHMED

3. A particle P is moving with constant acceleration $(-4\mathbf{i} + \mathbf{j})\text{ms}^{-2}$

At time $t = 0$, P has velocity $(14\mathbf{i} - 5\mathbf{j})\text{ms}^{-1}$

- (a) Find the speed of P at time $t = 2$ seconds.

(3)

DR ESAM AHMED

3. A particle P is moving with constant acceleration $(-4\mathbf{i} + \mathbf{j})\text{ms}^{-2}$

At time $t = 0$, P has velocity $(14\mathbf{i} - 5\mathbf{j})\text{ms}^{-1}$

- (b) Find the size of the angle between the direction of $\mathbf{i}$ and the direction of motion of P at time $t = 2$ seconds.

(3)

DR ESAM AHMED

3. A particle P is moving with constant acceleration $(-4\mathbf{i} + \mathbf{j})\text{ms}^{-2}$

At time $t = 0$, P has velocity $(14\mathbf{i} - 5\mathbf{j})\text{ms}^{-1}$

At time $t = T$ seconds, P is moving in the direction of vector $(2\mathbf{i} - 3\mathbf{j})$

(c) Find the value of T

(4)

DR ESAM AHMED

3. A particle moves on a smooth horizontal plane under the action of three horizontal forces, $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, where

$$\mathbf{F}_1 = (2p\mathbf{i} - 3p\mathbf{j})\text{N}$$

$$\mathbf{F}_2 = (q\mathbf{i} + \mathbf{j})\text{N}$$

$$\mathbf{F}_3 = (3\mathbf{i} - q\mathbf{j})\text{N}$$

where p and q are constants.

The resultant force acts in the direction of the vector $(4\mathbf{i} - 5\mathbf{j})$

- (a) Show that

$$2p - q = 19$$

(4)

DR ESAM AHMED

3. A particle moves on a smooth horizontal plane under the action of three horizontal forces, $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, where

$$\mathbf{F}_1 = (2p\mathbf{i} - 3p\mathbf{j})\text{N}$$

$$\mathbf{F}_2 = (q\mathbf{i} + \mathbf{j})\text{N}$$

$$\mathbf{F}_3 = (3\mathbf{i} - q\mathbf{j})\text{N}$$

where p and q are constants.

The mass of the particle is 0.5 kg.

Given that $p = 7$

- (b) find the acceleration of the particle.

(4)

DR ESAM AHMED

TOPIC 02 | FAMILY 03

Position vectors and relative motion

WHAT THIS FAMILY ASKS

Model one or more moving particles with $r = r_0 + vt$ and use relative position to answer meeting, direction or distance questions.

VARIANT ROUTE

V01 Meeting or route intersection

V02 Relative bearing or distance at a time

V03 Infer a landmark or route point

V04 Timed position with distance/angle locus

15 classified questions

186 total marks

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

At 12:00, a ship P sets sail from a harbour with position vector $(15\mathbf{i} + 36\mathbf{j})\text{km}$.

At 12:30, P is at the point with position vector $(20\mathbf{i} + 34\mathbf{j})\text{km}$.

Given that P moves with constant velocity,

- (a) show that the velocity of P is $(10\mathbf{i} - 4\mathbf{j})\text{km h}^{-1}$

(2)

DR ESAM AHMED

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

At 12:00, a ship P sets sail from a harbour with position vector $(15\mathbf{i} + 36\mathbf{j})\text{ km}$.

At 12:30, P is at the point with position vector $(20\mathbf{i} + 34\mathbf{j})\text{ km}$.

Given that P moves with constant velocity,

At time t hours after 12:00, the position vector of P is $\mathbf{p}\text{ km}$.

- (b) Find an expression for $\mathbf{p}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and t .

(2)

DR ESAM AHMED

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

A second ship Q is also travelling at a constant velocity.

At time t hours after 12:00, the position vector of Q is given by $\mathbf{q}$ km, where

$$\mathbf{q} = (42 - 8t)\mathbf{i} + (9 + 14t)\mathbf{j}$$

Ships P and Q are modelled as particles.

If both ships maintained their course,

- (c) (i) verify that they would collide at 13:30
 (ii) find the position vector of the point at which the collision would occur.

(4)

DR ESAM AHMED

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

At 12:00, a ship P sets sail from a harbour with position vector $(15\mathbf{i} + 36\mathbf{j})\text{ km}$.

At 12:30, P is at the point with position vector $(20\mathbf{i} + 34\mathbf{j})\text{ km}$.

Given that P moves with constant velocity,

A second ship Q is also travelling at a constant velocity.

At time t hours after 12:00, the position vector of Q is given by $\mathbf{q}$ km, where

$$\mathbf{q} = (42 - 8t)\mathbf{i} + (9 + 14t)\mathbf{j}$$

At 12:30 Q changes speed and direction to avoid the collision.

Ship Q now travels due north with a constant speed of 15 km h^{-1}

Ship P maintains the same constant velocity throughout.

- (d) Find the exact distance between P and Q at 14:30

(7)

DR ESAM AHMED

3. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors and position vectors are given relative to a fixed origin.]

A ship A is moving with constant velocity.

At 1 pm, the position vector of A is $(25\mathbf{i} + 10\mathbf{j})$ km.

At 3 pm, the position vector of A is $(55\mathbf{i} + 34\mathbf{j})$ km.

At time t hours after 1 pm, the position vector of A is $\mathbf{r}_A$ km.

- (a) Show that $\mathbf{r}_A = (25 + 15t)\mathbf{i} + (10 + 12t)\mathbf{j}$

(4)

DR ESAM AHMED

A ship A is moving with constant velocity.

At 1 pm, the position vector of A is $(25\mathbf{i} + 10\mathbf{j})$ km.

At 3 pm, the position vector of A is $(55\mathbf{i} + 34\mathbf{j})$ km.

The speed of A is $V\text{ms}^{-1}$

(b) Find the value of V .

(2)

DR ESAM AHMED

3. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors and position vectors are given relative to a fixed origin.]

A ship A is moving with constant velocity.

At 1 pm, the position vector of A is $(25\mathbf{i} + 10\mathbf{j})$ km.

At 3 pm, the position vector of A is $(55\mathbf{i} + 34\mathbf{j})$ km.

A ship B is moving with constant velocity $(20\mathbf{i} - 6\mathbf{j})$ km h⁻¹

At 1 pm, the position vector of B is $(35\mathbf{i} + 51\mathbf{j})$ km.

At 2:30 pm, B passes through the point P .

(c) Show that A also passes through P .

(5)

DR ESAM AHMED

8. [In this question, the horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two speedboats, A and B , are each moving with constant velocity. The velocity of A is 20 km h^{-1} due west and the velocity of B is 40 km h^{-1} on a bearing of 150° . The boats are modelled as particles.

At noon, the position vector of A is $60\mathbf{i} \text{ km}$ and B is at the origin O . At time t hours after noon, the position vector of A is $\mathbf{r} \text{ km}$ and the position vector of B is $\mathbf{s} \text{ km}$.

- (a) Find the velocity of B in the form $(p\mathbf{i} + q\mathbf{j}) \text{ km h}^{-1}$

(3)

DR ESAM AHMED

8. [In this question, the horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two speedboats, A and B , are each moving with constant velocity. The velocity of A is 20 km h^{-1} due west and the velocity of B is 40 km h^{-1} on a bearing of 150° . The boats are modelled as particles.

At noon, the position vector of A is $60\mathbf{i} \text{ km}$ and B is at the origin O . At time t hours after noon, the position vector of A is $\mathbf{r} \text{ km}$ and the position vector of B is $\mathbf{s} \text{ km}$.

- (b) Find expressions for $\mathbf{r}$ and $\mathbf{s}$ in terms of t , $\mathbf{i}$ and $\mathbf{j}$.

(3)

DR ESAM AHMED

8. [In this question, the horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two speedboats, A and B , are each moving with constant velocity. The velocity of A is 20 km h^{-1} due west and the velocity of B is 40 km h^{-1} on a bearing of 150° . The boats are modelled as particles.

At noon, the position vector of A is $60\mathbf{i}$ km and B is at the origin O . At time t hours after noon, the position vector of A is $\mathbf{r}$ km and the position vector of B is $\mathbf{s}$ km.

- (c) Find the time, to the nearest minute, at which the distance between the boats is the same as it was at noon.

(8)

DR ESAM AHMED

8. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.]

A ship A moves with constant velocity $(3\mathbf{i} - 10\mathbf{j})\text{ km h}^{-1}$

At time t hours, the position vector of A is $\mathbf{r}$ km.

At time $t = 0$, A is at the point with position vector $(13\mathbf{i} + 5\mathbf{j})\text{ km}$.

- (a) Find $\mathbf{r}$ in terms of t .

(2)

DR ESAM AHMED

8. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.]

A ship A moves with constant velocity $(3\mathbf{i} - 10\mathbf{j})\text{ km h}^{-1}$

At time t hours, the position vector of A is $\mathbf{r}$ km.

At time $t = 0$, A is at the point with position vector $(13\mathbf{i} + 5\mathbf{j})\text{ km}$.

Another ship B moves with constant velocity $(15\mathbf{i} + 14\mathbf{j})\text{ km h}^{-1}$

At time $t = 0$, B is at the point with position vector $(3\mathbf{i} - 5\mathbf{j})\text{ km}$.

(b) Show that, at time t hours,

$$\overrightarrow{AB} = [(12t - 10)\mathbf{i} + (24t - 10)\mathbf{j}] \text{ km} \quad (4)$$

DR ESAM AHMED

8. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.]

Given that the two ships do not change course,

- (c) find the shortest distance between the two ships,

(6)

DR ESAM AHMED

8. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.]
Given that the two ships do not change course,
(d) find the bearing of ship B from ship A when the ships are closest.

(2)

DR ESAM AHMED

8. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two boats, P and Q , are moving with constant velocities.

The velocity of P is $15\mathbf{i} \text{ ms}^{-1}$ and the velocity of Q is $(20\mathbf{i} - 20\mathbf{j}) \text{ ms}^{-1}$

- (a) Find the direction in which Q is travelling, giving your answer as a bearing.

(2)

DR ESAM AHMED

8. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two boats, P and Q , are moving with constant velocities.

The velocity of P is $15\mathbf{i} \text{ ms}^{-1}$ and the velocity of Q is $(20\mathbf{i} - 20\mathbf{j}) \text{ ms}^{-1}$

The boats are modelled as particles.

At time $t = 0$, P is at the origin O and Q is at the point with position vector $200\mathbf{j} \text{ m}$.

At time t seconds, the position vector of P is $\mathbf{p} \text{ m}$ and the position vector of Q is $\mathbf{q} \text{ m}$.

(b) Show that

$$\vec{PQ} = [5t\mathbf{i} + (200 - 20t)\mathbf{j}] \text{ m} \quad (5)$$

DR ESAM AHMED

8. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

$$\vec{PQ} = [5t\mathbf{i} + (200 - 20t)\mathbf{j}] \text{ m} \quad (5)$$

- (c) Find the bearing of P from Q when $t = 10$ (2)

DR ESAM AHMED

8. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

$$\vec{PQ} = [5t\mathbf{i} + (200 - 20t)\mathbf{j}] \text{ m} \quad (5)$$

- (d) Find the distance between P and Q when Q is north east of P (5)

DR ESAM AHMED

The boats are modelled as particles.

At time $t = 0$, P is at the origin O and Q is at the point with position vector $200\mathbf{j}$ m.

At time t seconds, the position vector of P is $\mathbf{p}$ m and the position vector of Q is $\mathbf{q}$ m.

$$\overrightarrow{PQ} = [5t\mathbf{i} + (200 - 20t)\mathbf{j}] \text{ m} \quad (5)$$

(e) Find the times when P and Q are 200 m apart.

(3)

DR ESAM AHMED

8. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two ships, A and B , are moving with constant velocities.

The velocity of A is $(3\mathbf{i} + 12\mathbf{j})\text{ km h}^{-1}$ and the velocity of B is $(p\mathbf{i} + q\mathbf{j})\text{ km h}^{-1}$

- (a) Find the speed of A .

(2)

DR ESAM AHMED

8. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two ships, A and B , are moving with constant velocities.

The velocity of A is $(3\mathbf{i} + 12\mathbf{j})\text{ km h}^{-1}$ and the velocity of B is $(p\mathbf{i} + q\mathbf{j})\text{ km h}^{-1}$

The ships are modelled as particles.

At 12 noon, A is at the point with position vector $(-9\mathbf{i} + 6\mathbf{j})\text{ km}$ and B is at the point with position vector $(16\mathbf{i} + 6\mathbf{j})\text{ km}$.

At time t hours after 12 noon,

$$\overrightarrow{AB} = [(25 - 12t)\mathbf{i} - 9t\mathbf{j}]\text{ km}$$

- (b) Find the value of p and the value of q .

(7)

DR ESAM AHMED

8. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

$$\overrightarrow{AB} = [(25 - 12t)\mathbf{i} - 9t\mathbf{j}] \text{ km}$$

- (c) Find the bearing of A from B when the ships are 15 km apart, giving your answer to the nearest degree.

(7)

DR ESAM AHMED

A square floor space $ABCD$, with centre O , is modelled as a flat horizontal surface measuring 50 m by 50 m, as shown in Figure 5.

The horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are in the direction of $\overrightarrow{AB}$ and $\overrightarrow{AD}$ respectively.

All position vectors are given relative to O .

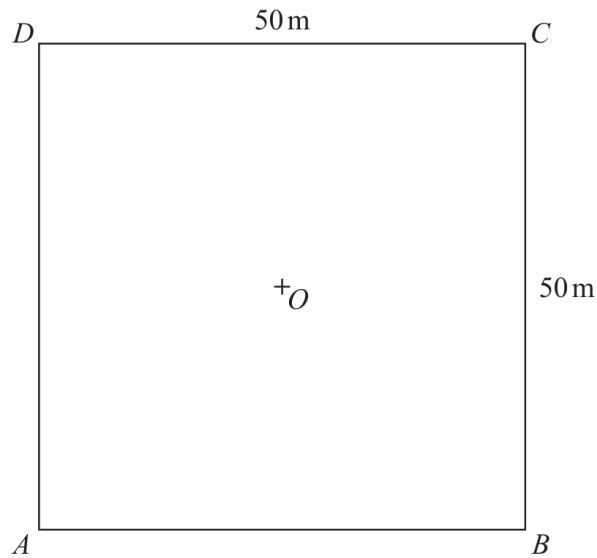
A small robot R is programmed to travel across the floor at a constant velocity.

- At time $t = 0$, R is at the point with position vector $(-2\mathbf{i} + \mathbf{j})\text{ m}$
- At time $t = 11\text{ s}$, R is at the point with position vector $(9\mathbf{i} + 23\mathbf{j})\text{ m}$
- At time t seconds, the position vector of R is $\mathbf{r}$ metres

(a) Find, in terms of t , $\mathbf{i}$ and $\mathbf{j}$, an expression for $\mathbf{r}$

(3)

DR ESAM AHMED



measuring 50 m by 50 m, as shown in Figure 5.

The horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are in the direction of $\vec{AB}$ and $\vec{AD}$ respectively.

All position vectors are given relative to O .

A second robot S is at the point C .

- At time $t = 0$, S leaves C and moves with constant velocity $(-\mathbf{i} - \mathbf{j}) \text{ m s}^{-1}$
- At time t seconds, the position vector of S is $\mathbf{s}$ metres

(b) Write down, in terms of t , $\mathbf{i}$ and $\mathbf{j}$, an expression for $\mathbf{s}$

(1)

(c) Show that

$$\overrightarrow{SR} = [(2t - 27)\mathbf{i} + (3t - 24)\mathbf{j}] \text{ m}$$

(2)

DR ESAM AHMED

$$\overrightarrow{SR} = [(2t - 27)\mathbf{i} + (3t - 24)\mathbf{j}] \text{ m}$$

(2)

(d) Find the time when the distance between R and S is a minimum.

(3)

DR ESAM AHMED

7. [In this question, the horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two speedboats, A and B , are each moving with constant velocity.

- the velocity of A is 40 km h^{-1} due east
- the velocity of B is 20 km h^{-1} on a bearing of angle α ($0^\circ < \alpha < 90^\circ$), where $\tan \alpha = \frac{4}{3}$

The boats are modelled as particles.

- (a) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the velocity of B in km h^{-1}

(2)

DR ESAM AHMED

7. [In this question, the horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two speedboats, A and B , are each moving with constant velocity.

- the velocity of A is 40 km h^{-1} due east
- the velocity of B is 20 km h^{-1} on a bearing of angle α ($0^\circ < \alpha < 90^\circ$), where $\tan \alpha = \frac{4}{3}$

The boats are modelled as particles.

At noon

- the position vector of A is $20\mathbf{j}$ km
- the position vector of B is $(10\mathbf{i} + 5\mathbf{j})$ km

At time t hours after noon

- the position vector of A is $\mathbf{r}$ km, where $\mathbf{r} = 20\mathbf{j} + 40t\mathbf{i}$
- the position vector of B is $\mathbf{s}$ km

- (b) Find an expression for $\mathbf{s}$ in terms of t , $\mathbf{i}$ and $\mathbf{j}$.

(2)

DR ESAM AHMED

7. [In this question, the horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

At time t hours after noon

- the position vector of A is $\mathbf{r}$ km, where $\mathbf{r} = 20\mathbf{j} + 40t\mathbf{i}$
- the position vector of B is $\mathbf{s}$ km

(c) Show that at time t hours after noon,

$$\overrightarrow{AB} = [(10 - 24t)\mathbf{i} + (12t - 15)\mathbf{j}] \text{ km}$$

(2)

DR ESAM AHMED

7. [In this question, the horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

$$\overrightarrow{AB} = [(10 - 24t)\mathbf{i} + (12t - 15)\mathbf{j}] \text{ km}$$

(2)

- (d) Show that the boats will never collide.

(3)

DR ESAM AHMED

7. [In this question, the horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

$$\overrightarrow{AB} = [(10 - 24t)\mathbf{i} + (12t - 15)\mathbf{j}] \text{ km}$$

(2)

- (e) Find the distance between the boats when the bearing of B from A is 225°

(4)

DR ESAM AHMED

5. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors and position vectors are given relative to a fixed origin.]

In a game, a ball B is rolled across a horizontal surface towards a fixed target.
The ball is modelled as a particle moving with constant velocity.

At time $t = 1$ s, the position vector of B is $(-2\mathbf{i} + 5\mathbf{j})$ m.

At time $t = 9$ s, the position vector of B is $(10\mathbf{i} - 3\mathbf{j})$ m.

- (a) Find the velocity of the ball.

(3)

DR ESAM AHMED

5. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors and position vectors are given relative to a fixed origin.]

In a game, a ball B is rolled across a horizontal surface towards a fixed target.
The ball is modelled as a particle moving with constant velocity.

At time $t = 1$ s, the position vector of B is $(-2\mathbf{i} + 5\mathbf{j})$ m.

At time $t = 9$ s, the position vector of B is $(10\mathbf{i} - 3\mathbf{j})$ m.

The position vector of the target is $(13\mathbf{i} - 5\mathbf{j})$ m.

- (b) Use the model to find the distance of B from the target at time $t = 7$ s.

(4)

DR ESAM AHMED

2. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively. Position vectors are given relative to a fixed origin.]

Two particles, A and B , are moving on a smooth horizontal surface.
Each particle is moving with constant velocity.

At time t seconds, the position vector of A is given by $\mathbf{r}$ metres.

- When $t = 2$, $\mathbf{r} = (-5\mathbf{i} + 16\mathbf{j})$
- When $t = 5$, $\mathbf{r} = (10\mathbf{i} + 4\mathbf{j})$

(a) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the velocity of A .

(2)

DR ESAM AHMED

2. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively. Position vectors are given relative to a fixed origin.]

Two particles, A and B , are moving on a smooth horizontal surface.
Each particle is moving with constant velocity.

At time t seconds, the position vector of A is given by $\mathbf{r}$ metres.

- When $t = 2$, $\mathbf{r} = (-5\mathbf{i} + 16\mathbf{j})$
 - When $t = 5$, $\mathbf{r} = (10\mathbf{i} + 4\mathbf{j})$
- (b) Find an expression for $\mathbf{r}$ at time t seconds.
Give your answer in the form $p\mathbf{i} + q\mathbf{j}$, where p and q are functions of t

(2)

DR ESAM AHMED

2. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively. Position vectors are given relative to a fixed origin.]

At time t seconds, the position vector of A is given by $\mathbf{r}$ metres.

- When $t = 2$, $\mathbf{r} = (-5\mathbf{i} + 16\mathbf{j})$
- When $t = 5$, $\mathbf{r} = (10\mathbf{i} + 4\mathbf{j})$

At time t seconds, the position vector of B is given by $\mathbf{s}$ metres where

$$\mathbf{s} = -5\mathbf{i} + 7\mathbf{j} + t(2\mathbf{i} - 3\mathbf{j})$$

- (c) Find, to the nearest degree, the bearing of B from A when $t = 5$

(3)

DR ESAM AHMED

6. [In this question **i** and **j** are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin *O*.]

At 9 am, a ship *S* leaves a harbour and moves with constant velocity $(10\mathbf{i} + 15\mathbf{j})\text{km h}^{-1}$

(a) Find the exact speed of *S*

(2)

DR ESAM AHMED

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

The harbour is at the origin O .

At 9 am, a boat B leaves the point with position vector $(50\mathbf{i} + 10\mathbf{j})\text{km}$ and moves with constant velocity $(-8\mathbf{i} + 10\mathbf{j})\text{km h}^{-1}$

- (b) Find the direction in which B is moving, giving your answer as a bearing to the nearest degree.

(3)

DR ESAM AHMED

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

At 9 am, a ship S leaves a harbour and moves with constant velocity $(10\mathbf{i} + 15\mathbf{j})\text{km h}^{-1}$.
The harbour is at the origin O .

At 9 am, a boat B leaves the point with position vector $(50\mathbf{i} + 10\mathbf{j})\text{km}$ and moves with constant velocity $(-8\mathbf{i} + 10\mathbf{j})\text{km h}^{-1}$.

(c) Show that at time t hours after 9 am,

$$\overrightarrow{SB} = [(50 - 18t)\mathbf{i} + (10 - 5t)\mathbf{j}]\text{km}$$

(4)

DR ESAM AHMED

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

$$\overrightarrow{SB} = [(50 - 18t)\mathbf{i} + (10 - 5t)\mathbf{j}] \text{ km}$$

(4)

- (d) Show that B and S do not collide.

(3)

DR ESAM AHMED

ELITE EDITORIAL CORRECTION — SOURCE-BACKED

In part (e), read “south-west” instead of the printed “north-east”.

Why: the final mark scheme flags an SW/NE error. Equating the i - and j -components gives $t = 40/13$ and $\vec{SB} = -(70/13)\mathbf{i} - (70/13)\mathbf{j}$, so B is south-west of S .

The original Pearson question pixels are preserved unchanged below.

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

$$\vec{SB} = [(50 - 18t)\mathbf{i} + (10 - 5t)\mathbf{j}] \text{ km}$$

(4)

When B is north-east of S , the distance between S and B is d km.

- (e) Find the value of d .

(4)

DR ESAM AHMED

7. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

A boat, which is modelled as a particle, is moving with constant velocity $(-2\mathbf{i} + \mathbf{j}) \text{ km h}^{-1}$

At time $t = 1.5$ hours after noon, the boat passes through the point with position vector $(11\mathbf{i} + 3.5\mathbf{j}) \text{ km}$.

- (a) Show that, at time t hours after noon, the position vector of the boat is

$$[(14 - 2t)\mathbf{i} + (2 + t)\mathbf{j}] \text{ km}$$

(3)

DR ESAM AHMED

7. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

A boat, which is modelled as a particle, is moving with constant velocity $(-2\mathbf{i} + \mathbf{j}) \text{ km h}^{-1}$

At time $t = 1.5$ hours after noon, the boat passes through the point with position vector $(11\mathbf{i} + 3.5\mathbf{j}) \text{ km}$.

The boat is visible from O when it is 9 km or less from O .

- (b) Find the length of time for which the boat is visible from O , showing your method clearly.

(4)

DR ESAM AHMED

7. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

$$[(14 - 2t)\mathbf{i} + (2 + t)\mathbf{j}] \text{ km}$$

(3)

- (c) Find the value of t when the boat is northwest of O .

(2)

DR ESLAM AHMED

7. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

A boat, which is modelled as a particle, is moving with constant velocity $(-2\mathbf{i} + \mathbf{j}) \text{ km h}^{-1}$

At the instant when the boat is at the point P , OP is perpendicular to the direction in which the boat is moving.

- (d) Show that $OP = \frac{18\sqrt{5}}{5} \text{ km}$

(Solutions relying on calculator technology are not acceptable.)

(6)

DR ESAM AHMED

6. Two girls, Agatha and Brionie, are roller skating inside a large empty building. The girls are modelled as particles.

At time $t = 0$, Agatha is at the point with position vector $(11\mathbf{i} + 11\mathbf{j})\text{m}$ and Brionie is at the point with position vector $(7\mathbf{i} + 16\mathbf{j})\text{m}$. The position vectors are given relative to the door, O , and $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.

Agatha skates with constant velocity $(3\mathbf{i} - \mathbf{j})\text{ms}^{-1}$

Brionie skates with constant velocity $(4\mathbf{i} - 2\mathbf{j})\text{ms}^{-1}$

- (a) Find the position vector of Agatha at time t seconds.

(2)

DR ESAM AHMED

6. Two girls, Agatha and Brionie, are roller skating inside a large empty building. The girls are modelled as particles.

At time $t = 0$, Agatha is at the point with position vector $(11\mathbf{i} + 11\mathbf{j})\text{m}$ and Brionie is at the point with position vector $(7\mathbf{i} + 16\mathbf{j})\text{m}$. The position vectors are given relative to the door, O , and $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.

Agatha skates with constant velocity $(3\mathbf{i} - \mathbf{j})\text{ms}^{-1}$

Brionie skates with constant velocity $(4\mathbf{i} - 2\mathbf{j})\text{ms}^{-1}$

At time $t = 6$ seconds, Agatha passes through the point P .

- (b) Show that Brionie also passes through P and find the value of t when this occurs.

(4)

DR ESAM AHMED

6. Two girls, Agatha and Brionie, are roller skating inside a large empty building. The girls are modelled as particles.

At time $t = 0$, Agatha is at the point with position vector $(11\mathbf{i} + 11\mathbf{j})\text{m}$ and Brionie is at the point with position vector $(7\mathbf{i} + 16\mathbf{j})\text{m}$. The position vectors are given relative to the door, O , and $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.

Agatha skates with constant velocity $(3\mathbf{i} - \mathbf{j})\text{ms}^{-1}$

Brionie skates with constant velocity $(4\mathbf{i} - 2\mathbf{j})\text{ms}^{-1}$

At time t seconds, Agatha is at the point A and Brionie is at the point B .

(c) Show that $\overrightarrow{AB} = [(t - 4)\mathbf{i} + (5 - t)\mathbf{j}]\text{m}$

(2)

DR ESAM AHMED

6. Two girls, Agatha and Brionie, are roller skating inside a large empty building. The girls are modelled as particles.

(c) Show that $\overrightarrow{AB} = [(t - 4)\mathbf{i} + (5 - t)\mathbf{j}] \text{ m}$ (2)

- (d) Find the distance between the two girls when they are closest together. (4)

DR ESAM AHMED

8. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.]

At 7 am a ship leaves a port and moves with constant velocity. The position vector of the port is $(-2\mathbf{i} + 9\mathbf{j})$ km.

At 7.36 am the ship is at the point with position vector $(4\mathbf{i} + 6\mathbf{j})$ km.

- (a) Show that the velocity of the ship is $(10\mathbf{i} - 5\mathbf{j}) \text{ km h}^{-1}$

(2)

DR ESAM AHMED

8. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.]

At 7 am a ship leaves a port and moves with constant velocity. The position vector of the port is $(-2\mathbf{i} + 9\mathbf{j})$ km.

At 7.36 am the ship is at the point with position vector $(4\mathbf{i} + 6\mathbf{j})$ km.

(b) Find the position vector of the ship t hours after leaving port.

(2)

DR ESAM AHMED

8. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.]

At 8.48 am a passenger on the ship notices that a lighthouse is due east of the ship.

At 9 am the same passenger notices that the lighthouse is now north east of the ship.

- (c) Find the position vector of the lighthouse.

(4)

DR ESLAM AHMED

8. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin.]

At 8.48 am a passenger on the ship notices that a lighthouse is due east of the ship.

At 9 am the same passenger notices that the lighthouse is now north east of the ship.

(d) Find the position vector of the ship when it is due south of the lighthouse.

(4)

DR ESAM AHMED

4. The position vector, $\mathbf{r}$ metres, of a particle P at time t seconds, relative to a fixed origin O , is given by

$$\mathbf{r} = (t - 3)\mathbf{i} + (1 - 2t)\mathbf{j}$$

- (a) Find, to the nearest degree, the size of the angle between $\mathbf{r}$ and the vector $\mathbf{j}$, when $t = 2$
(3)

DR ESLAM AHMED

4. The position vector, $\mathbf{r}$ metres, of a particle P at time t seconds, relative to a fixed origin O , is given by

$$\mathbf{r} = (t - 3)\mathbf{i} + (1 - 2t)\mathbf{j}$$

- (b) Find the values of t for which the distance of P from O is 2.5 m.

(5)

DR ESAM AHMED

TOPIC 02 | FAMILY 04

Closest approach in relative motion

WHAT THIS FAMILY ASKS

Minimise the separation of particles moving at constant vector velocities.

VARIANT ROUTE

V01 Minimise distance squared

1 classified questions

11 total marks

7. [In this question **i** and **j** are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin *O*.]

At midnight, a ship *S* is at the point with position vector $(19\mathbf{i} + 22\mathbf{j})\text{ km}$

The ship travels with constant velocity $(12\mathbf{i} - 16\mathbf{j})\text{ km h}^{-1}$

(a) Find the speed of *S*.

(2)

DR ESAM AHMED

7. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

At midnight, a ship S is at the point with position vector $(19\mathbf{i} + 22\mathbf{j})\text{ km}$

The ship travels with constant velocity $(12\mathbf{i} - 16\mathbf{j})\text{ km h}^{-1}$

At time t hours after midnight, the position vector of S is $\mathbf{s}\text{ km}$.

- (b) Find an expression for $\mathbf{s}$ in terms of $\mathbf{i}$, $\mathbf{j}$ and t .

(2)

DR ESAM AHMED

7. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

At midnight, a ship S is at the point with position vector $(19\mathbf{i} + 22\mathbf{j})\text{ km}$

The ship travels with constant velocity $(12\mathbf{i} - 16\mathbf{j})\text{ km h}^{-1}$

A lighthouse stands on a small rocky island. The lighthouse is modelled as being at the point with position vector $(26\mathbf{i} + 15\mathbf{j})\text{ km}$.

It is not safe for ships to be within 1.3 km of the lighthouse.

- (c) (i) Find the value of t when S is closest to the lighthouse.

- (ii) Hence determine whether it is safe for S to continue its course.

(7)

DR ESAM AHMED